



Full length article

An exploration of urban air health navigation system based on dynamic exposure risk forecast of ambient PM2.5

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ABSTRACT

Under international advocacy for a low-carbon and healthy lifestyle, ambient PM2.5 pollution poses a dilemma for urban residents who wish to engage in outdoor exercise and adopt active low-carbon commuting. In this study, an Urban Air Health Navigation System (UAHNS) was designed and proposed to assist users by recommending routes with the least PM2.5 exposure and dynamically issuing early risk warnings based on topologized digital maps, an application programming interface (API), an eXtreme Gradient Boosting (XGBoost) model, and two-step spatial interpolation. A test of the UAHNS's functions and applications was carried out in Wuhan city. The results showed that, compared with trained random forest (RF), LightGBM, Adaboost models, etc., the XGBoost model performed better, with an R^2 exceeding 0.90 and an RMSE of approximately $15.74 \mu\text{g}/\text{m}^3$, based on data from national air and meteorological monitoring stations. Further, the two-step spatial interpolation model was adopted to dynamically generate pollution distribution at a spatial resolution of $300 \text{ m} \times 300 \text{ m}$. Then, an exposure comparison was performed under randomly selected commuting routes and times in Wuhan, showing the recommended routes for lower PM2.5 exposure made effectively help. And the route difference ratios of about 14.9 % and 16.9 % for riding and walking, respectively. Finally, the UAHNS platform was integrally realized for Wuhan, consisting of a real-time PM2.5 query, a one-hour PM2.5 prediction function at any location, health navigation on city map, and a personalized health information query.

1. Introduction

At present, more than half of the population around the globe resides in urbanized areas, where anthropogenic activities and emissions are heavily concentrated, causing an imbalanced distribution of emissions, as well as ecosystem degradation, decreased biodiversity, and a significant increase in public health risks (Motesharrei et al., 2016; Li et al., 2021). Common ambient air pollution issues in developing countries, such as PM2.5 pollution, a major contributor to haze, diminish social productivity and residential life quality (Yang et al., 2024). PM2.5 refers to fine inhalable atmospheric particles that float in the air with an aerodynamic diameter less than $2.5 \mu\text{m}$. Epidemiological studies have unequivocally demonstrated that PM2.5 concentrations result in an increased risk of cardiopulmonary and lung cancer (Shang et al., 2024; GBD, 2019 Risk Factors Collaborators, 2020). Exposure to ambient PM2.5 has been estimated to have caused more than 4.58 million global

deaths in 2017 (Bu et al., 2021). Among the primary sources of PM2.5, traffic emission is one of the important, and it is also a significant source of urban carbon emissions. Therefore, PM2.5 is relatively concentrated on roads and roadsides (Jareemit et al., 2023; Yang et al., 2022). Residents might be exposed to relatively high concentrations of PM2.5 during outdoor commuting and exercise (Ilenič et al., 2024).

To reduce residents' health risks and mitigate traffic pollutant emissions and the corresponding carbon emissions, numerous governments advocate for the adoption of green commuting methods, such as walking and cycling (Bigazzi and Figliozzi, 2015; Ministry of Transport of the People's Republic of China, 2023). In recent years, the proportion of active commuting (walking and riding) in urban traffic has gradually increased (Wang et al., 2020). A study in China found that 50 % of the sample residents already engaged in active commuting (Fan et al., 2019). Walking was common for short distances between metro stations and residential areas. It was also found that, in areas with limited public

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Table 1
The identified input features to XGBoost models.

Category	Characteristic	Characteristic explanation
Air pollutants	PM2.5 (μg/m ³)	PM2.5 concentration
	PM10 (μg/m ³)	PM10 concentration
	NO ₂ (μg/m ³)	NO ₂ concentration
	SO ₂ (μg/m ³)	SO ₂ concentration
	CO (μg/m ³)	CO concentration
	O ₃ (μg/m ³)	O ₃ concentration
Meteorological factors	Temp (°C)	Temperature
	Rel_Hum (%)	Relative humidity
	Pressure (Hpa)	Atmospheric pressure
	Wind_S (km/h)	Rate of air motion
	Precip (mm)	Rainfall amount
	Wind_D	Direction of wind
	Temp (°C)	Temperature

transport coverage, many residents chose to commute medium or long distances by electric bicycle, which is quite common in developing countries (Zabiulla et al., 2024). According to statistics, the sales of electric bicycles in China will reach 41 million, with a social population of about 330 million, ranking first in the world (Shanghai Airui Market Consulting Co., Ltd. 2022). Unfortunately, the above active commuting methods necessitate a higher respiration rate (Table A1 in Appendix) and more direct exposure to ambient air pollutants, potentially resulting in their elevated intake (Chen et al., 2023b; Wu et al., 2023). Therefore, it has become imperative to effectively guide residents in minimizing health risks while embracing a green commuting lifestyle. A previous study indicated that PM2.5 are unevenly distributed within a city, so there can be more than a 26 % difference in PM2.5 exposure when cycling in different areas within the same city (Tainio et al., 2021). Since the selection of commuting routes can greatly impact pollution exposure levels, there is a practical need for an intelligent and automatic route selection system that is integrated with an accurate temporal-spatial exposure risk assessment. In recent years, research on outdoor PM2.5 and health risks has gradually been upgraded from focusing on pollution concentrations at sites to individual exposure differences and their state of citizen environmental equity (Ou et al., 2024). The precise identification and management of commuting PM2.5 exposure would help reduce public health risks.

Previous research efforts have primarily concentrated on either the development of PM2.5 pollution level forecasting models (Bae et al., 2020; Zhang et al., 2018; Zhou et al., 2024) or the creation of a route planning system for various purposes (D’Emidio et al., 2024; Fernandes et al., 2024), but relatively few studies have integrated both into one intelligent application for ambient air health guidance and early warnings. In recent years, the number of machine learning methods used for forecasting PM2.5 has notably increased. Osowski and Garanty (2007) used support vector machines (SVMs) and wavelet decomposition for prediction based on meteorological data and pollutant concentrations. Further, Wang et al. (2017) used a hybrid linear and nonlinear approach to optimize a SVM model. Additionally, Huang et al. (2018) used aerosol optical depth, meteorological parameters, land cover, and ground PM2.5 measurements in the North China Plain from 2013 to 2015 to establish a random forest prediction model. And the overall R² and relative prediction error (RPE) of the model were 0.88 and 18.7 %. Kang et al. (2023) compared machine learning algorithms, such as RF, XGBoost, and LightGBM, using hourly meteorological and air quality data for PM2.5 forecasting, showing that the XGBoost model obtained optimal results for short-term predictions.

Regarding the issue of spatial interpolation methods, numerous scholars have conducted explorations, with Kriging and Inverse Distance Weighting (IDW) remaining the most classic interpolation techniques (Shukla et al., 2020). However, when there are insufficient sample points, a greater reliance on additional data is required for the basis of interpolation. The geographical environmental context is a significant factor (Zhao et al., 2020). For instance, Chen integrated land use with

interpolation methods to achieve certain research findings (Chen et al., 2019). The most referential work for us is the two-step interpolation method proposed by Zhang (2023), which involves initially expanding the data based on environmental conditions before conducting IDW interpolation. Furthermore, the essence of route planning is to identify the path with the lowest costs in the network diagram, which is equivalent to the shortest path problem in the graph data algorithm represented by the Dijkstra algorithm (Dijkstra, 1959). The development of the NoSQL database represented by Neo4j has shed light on the future direction of graph data algorithms. A large amount of road segment data can be easily stored and queried with a NoSQL database. A previous study showed that Neo4j is much suitable for storing and querying road segment network data, with higher efficiency in querying a route with the lowest cost under a shorter query time, making it as suitable database for route planning modules (Miler et al., 2014).

In this study, a user-oriented Urban Air Health Navigation System (UAHNS) is designed for active commuters and proposed to assist users by recommending the route with the lowest ambient PM2.5 exposure and dynamically issuing early risk warnings. Two main functional modules are included in the UAHNS: (1) The first one is a short-term PM2.5 forecast system, which is established using a trained XGBoost model based on hourly meteorological and air pollutant monitoring data. With this forecast system, prominently displayed on the homepage of the UAHNS, users can query the PM2.5 concentration in each location in the city on a map, calculated using the two-step spatial interpolation method. Furthermore, it combines the XGBoost algorithm to provide a predicted PM2.5 value for the next hour. (2) The second is an urban air health navigation module. This navigation module is built using the Django framework. Urban roads are abstracted as network diagrams, and these networks are transformed into nodes and relationships stored in the Neo4j graph database. Regarding the data structure of the above, this module implements a route planning algorithm to calculate and display the route with the least exposure. Finally, Wuhan is selected as an example to realize, test, and verify the UAHNS.

2. Data preparation

2.1. Data preprocessing for PM2.5 prediction

PM2.5 concentration is affected by various factors, mainly including meteorological and other air pollutant monitoring data (Chen et al., 2023a; Hou et al., 2022). In this study, PM2.5 prediction was conducted in Wuhan. Hourly meteorological and air pollutant monitoring data in Wuhan from January 1, 2023, to December 31, 2023, were obtained from the Wuhan public data open platform (<https://data.wuhan.gov.cn/>) and meteorological data website (<https://www.meteomanz.com/>). There has been a lot of research on the relationship between PM2.5, other air pollutants, and meteorological factors (Cui et al., 2023; Gao et al., 2024; Wei et al., 2021). Combined with our obtained data and the published research conclusions (Chen et al., 2020; Ou et al., 2022), the 6 air pollutant monitoring data (PM2.5, PM10, SO₂, NO₂, O₃, CO) and 6 meteorological factors (temperature, relative humidity, atmospheric pressure, wind speed, wind direction and precipitation) were finally adopted as the input features of the XGBoost model, as shown in Table 1. Data preprocessing addressed missing values via imputation with the mean and included feature normalization.

2.2. Data preprocessing for mapping of PM2.5 distribution

In our research, the spatial interpolation of PM2.5 was conducted by utilizing information from monitoring stations and certain geographical environmental data. The coordinates of the PM2.5 monitoring stations were derived from the official website of the Wuhan Municipal Ecology and Environment Bureau (<https://hbj.wuhan.gov.cn/>), which includes a total of 23 monitoring stations with their respective names, identification numbers, and geographic coordinates. The geographical

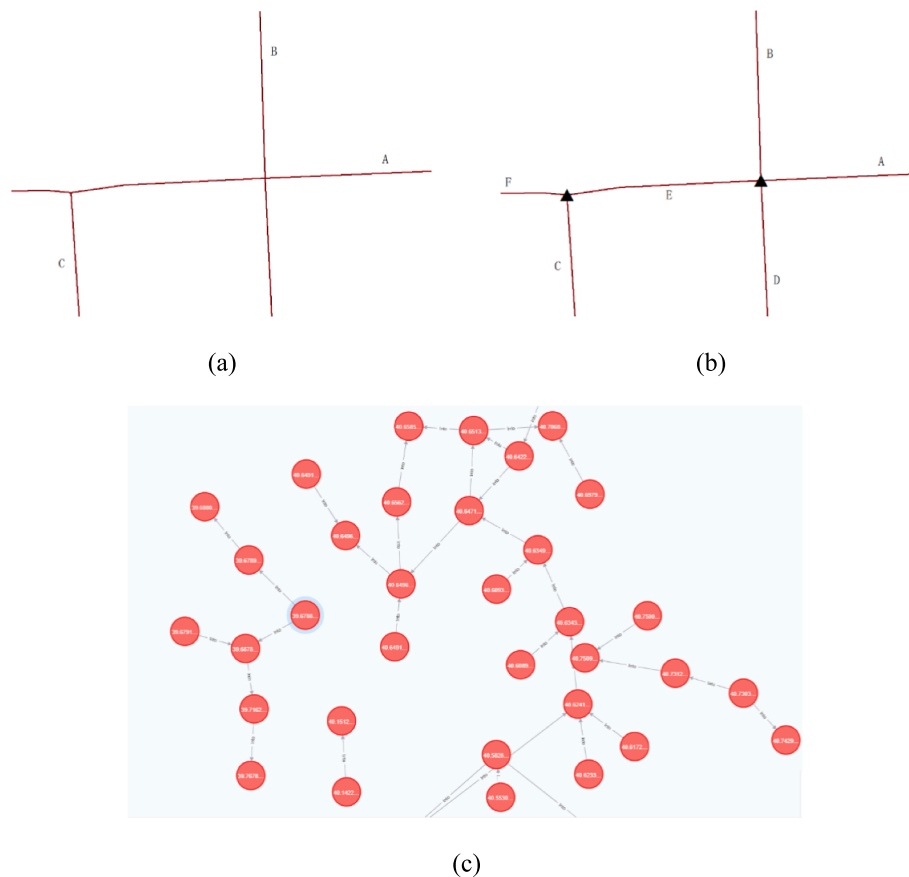


Fig. 1. Demonstration of split lines using planarize lines tool in ArcGIS: (a) before line split, (b) after line split, and (c) visualized result of road network stored in Neo4j graph database. Red dots represent the topologized intersection of road network segments and contain the IDs of the intersections and their GPS locations. Lines connecting each dot represent the relationship between the dots, which store the weights (calculated exposure level to PM_{2.5}) of the route and node IDs on both ends of the edge. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

environmental data encompass land use data obtained from the China Land Cover Data 30 m grid for Wuhan (Yang and Huang, 2021); population density data at a 1000 m grid resolution (2020) were obtained from WorldPop (<https://hub.worldpop.org/>); and traffic network data in Wuhan (2023) were sourced from Open Street Map (<https://www.openstreetmap.org/>). These data supply an important basis for our used environmental-similarity-based data enhancement technique, which will be detailedly introduced in section 3.3, and our spatial interpolation model.

2.3. Data preprocessing for digital map

Digitized maps are required for route searching and planning. The digital maps were downloaded from OpenStreetMap, which is an open-source map data platform (<https://www.openstreetmap.org/>). Digital maps are usually stored in vector format, containing the actual geographic data and information on various properties. The digital maps in this study were stored in Neo4j, which is a graph-network-oriented graph database developed using the Java language, and its data structure is based on graph theory. In the present study, the geographic data in the digital maps were abstracted into nodes and relationships in the Neo4j graph database for route planning using a topology method. Moreover, the road network segments were defined as relationships between nodes, while the intersections of the road segments were defined as the nodes. The digital map was topologized using ArcGIS software. Then, the planarize lines tool was used to split lines at intersections to build the topology structure of the road network. Fig. 1 (a) and 1 (b) show the principle of extracting routes and point features. Fig. 1 (c) presents an example of the visualized result of the road

network stored in the Neo4j graph database.

3. Functional models

3.1. Urban air health navigation system framework

The UAHNS is an urban route planning system based on identifying the route with the least PM_{2.5} exposure, and it is designed for daily use by citizens who use active commuting methods to help reduce carbon emissions, such as biking, electric bicycle, and walking. It consists of four functional modules and a web-based user application. The first functional model is a short-term PM_{2.5} concentration prediction model applying an XGBoost algorithm with the mentioned meteorological and air pollutant monitoring data. Users can navigate in real time or plan their trip in advance, but this functional model is only used for the latter. If users wish to navigate in real time, current data from weather stations will be pulled and directly fed into next functional module.

The second functional model spatially interpolates PM_{2.5} with the predicted PM_{2.5} value of each air quality monitoring station to obtain a fine-resolution predicted grid distribution of PM_{2.5} based on two-step spatial interpolation, which is introduced in detail in Section 3.3. The third functional model links the predicted grid distribution of PM_{2.5} with road network segments to establish a quantitative relative PM_{2.5} exposure model, and the relative exposure level of each road network segment is computed using the established model. The weight of each road segment is updated in the Neo4j graph database according to the relative exposure level.

The final functional model selects the route with the least exposure by using the graph algorithm built within Neo4j and the updated weight

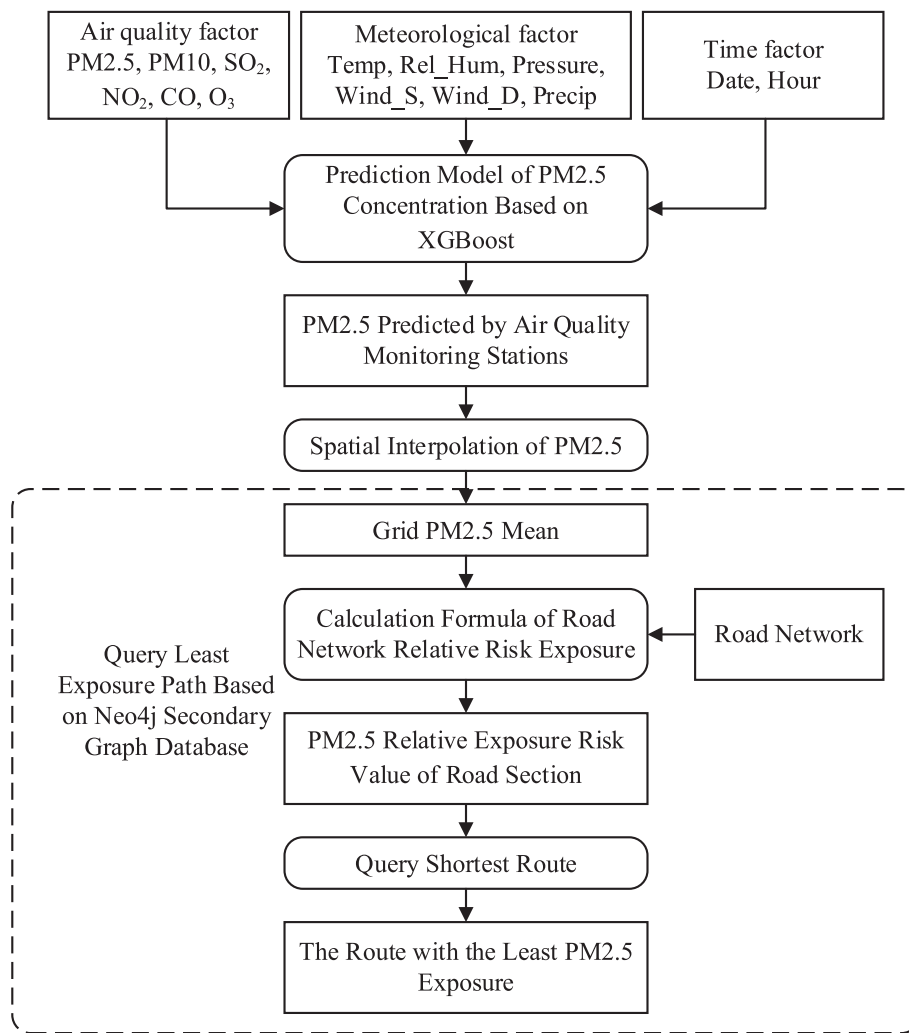


Fig. 2. The overall framework of the proposed UAHNS.

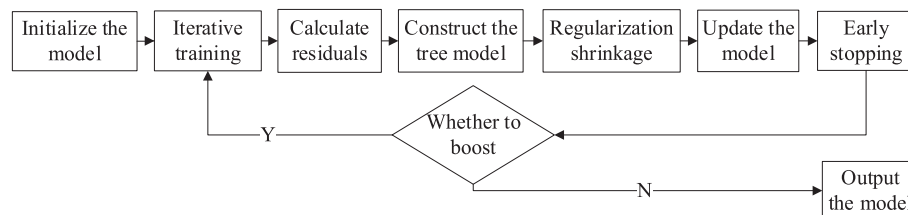


Fig. 3. Workflow of XGBoost.

attributes between nodes. A web-based user application was designed to integrate the four functional models and provide a user interface. It was built using the Django framework and was designed to meet basic functional and high-level user experience requirements. The overall framework of the proposed UAHNS is shown in Fig. 2.

3.2. PM2.5 prediction based on XGBoost

In this study, an XGBoost model was utilized to predict the concentration of PM2.5. XGBoost is an efficient ensemble learning algorithm based on the Gradient Boosting Decision Tree framework, which employs various optimization techniques to enhance the model's performance. Compared with traditional gradient boosting algorithms, XGBoost improves the model's predictive accuracy by progressively

adding new decision trees to correct the residuals from the previous round of iteration (Kang et al., 2020). A notable feature of XGBoost is the inclusion of regularization terms in the loss function, comprising both L1 and L2 regularizations, which help to reduce the model's complexity and effectively prevent overfitting. During the tree growth process, XGBoost adopts a depth-first strategy, where each split is based on the optimal feature and split point selected according to the information gain, until a preset maximum depth is reached or the amount of data within a node falls below a certain threshold. Moreover, XGBoost can properly handle missing values in training data, automatically choosing the best split path for these missing values. The algorithm's parallelized design significantly improves the model's training speed, and its ability to learn across features also enhances the model's capability to capture complex relationships. Several published studies (Huang, 2021; Kang et al.,

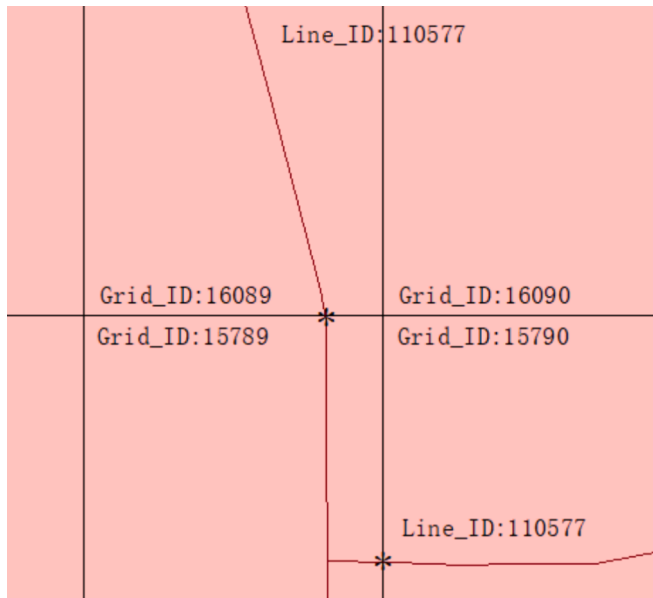


Fig. 4. Example of a road segment that needs to be separated.

2023) compared the short-term predictive ability of PM2.5 concentrations under common influencing factor set using mainstream machine learning algorithms (including Light Gradient Boosting Machine (LightGBM), XGBoost, Gradient Boosting Decision Tree (GBDT), Adaptive Boosting (AdaBoost) and RF), and it was found that XGBoost ran much efficiently on an hourly scale and owned good R^2 exceeding 0.87. Thus, an XGBoost model was adopted as the predictive model in this study.

The workflow of the XGBoost (Fig. 3) classifier begins with the initialization of the model, where a baseline prediction value, such as zero, is typically used as the starting point. Subsequently, the algorithm enters the iterative training phase, where it progressively refines the model through multiple iterations, constructing a decision tree in each round. In each iteration, the algorithm calculates the residuals between the current model's predictions and the actual target values. These residuals are then used as the training target for the new decision tree. The newly constructed decision tree model is optimized through a regularization and reduction process, where the learning rate (shrinkage) is used to adjust the contribution of the new tree to the model's predictions, and regularization terms are added to the loss function to prevent overfitting. The updated model integrates the contribution of the new tree to enhance the accuracy of predictions. After each iteration, the algorithm performs an early stopping check, evaluating the performance of the new tree through cross-validation. If the new tree fails to improve the model's performance, then the iteration process will be terminated. Ultimately, after several rounds of iteration and optimization, the algorithm outputs the trained XGBoost model, which is used for classifying and predicting new data.

To evaluate the performance of the trained XGBoost model, we used the RMSE, MAE, and R^2 to measure the fitting condition between the predicted value and label value. The equations for the performance measurements are as follows:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2} \quad (1)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i| \quad (2)$$

$$R^2 = \frac{\sum_{i=1}^n (Y_i - \bar{Y})(\hat{Y}_i - \bar{\hat{Y}})}{\sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2} \sqrt{\sum_{i=1}^n (\hat{Y}_i - \bar{\hat{Y}})^2}} \quad (3)$$

where Y_i represents the labeled value of the i_{th} sample, $\hat{Y}_i$ represents the predicted value of the i_{th} sample, $\bar{Y}$ represents the labeled average value, $\bar{\hat{Y}}$ represents the predicted average value, and n represents the total sample number.

3.3. PM2.5 spatial interpolation based on two-step method

In conventional spatial interpolation methods, the scarcity of known data can lead to issues such as insufficient detail and significant errors in the interpolation results. To achieve superior spatial interpolation outcomes, we aimed to alter the quantity of known data through certain means. When examining the known PM2.5 data, we observed that the relative severity of the pollution at various stations exhibited a degree of stability across different temporal cross-sections. In essence, certain stations consistently exhibited higher pollution levels than others, and vice versa. And these statistical findings could be found in Fig. A2 in Appendix. This temporal stability across different time periods suggests that the PM2.5 monitoring station results possess spatial correlation to a certain extent, which aligns with the Third Law of Geography (Zhu et al., 2018). Building upon this theoretical foundation, we proposed to predict the PM2.5 levels at some unknown points based on the degree of geographical environmental similarity. This approach was first employed by Zhang (2023), and a detailed explanation of the computational process was provided in the following paragraph. Data augmentation based on environmental similarity was the first step in our interpolation process. Once the results were obtained, we then proceeded with IDW interpolation on a more abundant dataset, thereby achieving enhanced spatial interpolation results. As two rounds of predictive interpolation are performed, this method could be called as the "two-step interpolation method based on environmental similarity," abbreviated as the two-step interpolation method.

Among all 23 environmental monitoring stations, the 2 most northern ones were distinct "outliers" and covered little of the population. Hence, after their exclusion, the study area was effectively concentrated, and the relative density of the monitoring points increased by 73 %. Our revised study area, a rectangle that contains Wuhan's central and southern areas, including more than 95 % of residents, uses 1,200 sampling points spaced 3 km apart. By analyzing 300 m, 900 m, and 1,500 m buffer zones around these points for environmental features, we derived an 18-dimensional vector. This vector aided in identifying points with the desired environmental similarity. During preprocessing, 1,200 environmental feature vectors were obtained, any one of them was in the shape of $e = (e_1, e_2, e_3, \dots, e_{18})$. Initially, 21 environmental monitoring stations were paired with point i via the nearest distance method, resulting in 1,179 unknown points and 21 known points. The environmental similarity calculation method is as follows:

$$S_{ij} = P(E(e_{i1}, e_{j1}), E(e_{i2}, e_{j2}), \dots, E(e_{i18}, e_{j18})) \quad (4)$$

$$E(e_{iv}, e_{jv}) = \exp \left(- \frac{(e_{iv} - e_{jv})^2}{2 \left(SD_{e_v} \times \frac{SD_{e_v}}{SD_{e_{jv}}} \right)^2} \right) \quad (5)$$

$$SD_{e_{jv}} = \sqrt{\frac{\sum_{i=1}^k (e_{iv} - e_{jv})^2}{k}} \quad (6)$$

where e_{iv} represents the v -th dimension of the vector for the i -th point, and SD stands for standard deviation. By applying the minimum operator method based on Liebig's Law of the Minimum (Van der Ploeg et al., 1999), we solved P to obtain an environmental similarity vector for each point of interest.

Subsequently, we calculated the environmental similarity of the 1,179 unknown points to the 21 sample points, selecting 99 sampling

points with an environmental similarity of over 0.85. We then estimated the PM_{2.5} values for these points. This approach allowed us to expand the limited original data of 21 points to 120 sample points suitable for subsequent interpolation, without incurring excessive computational costs. After obtaining these data, we then plotted the IDW interpolation results at a resolution of 300 m* 300 m. The formula for IDW is as follows:

$$\hat{Z}(s_0) = \sum_{i=1}^N \lambda_i Z(s_i) \quad (7)$$

where $\hat{Z}(s_0)$ represents the prediction value of an unmeasured position, N is the total number of measuring positions in study area, λ_i is the distance weight of the measured value of the i_{th} position, and $Z(s_i)$ is the measured value of the i_{th} position.

3.4. Selection of route with the least PM_{2.5} exposure

Route selection in the UAHNS is based on the relative PM_{2.5} exposure level (Sharker and Karimi, 2014). After the IDW areal interpolation, each road network segment's relative exposure level to PM_{2.5} can be calculated based on the predicted PM_{2.5} value within the grid cell that its couples with. However, it is common for one road network segment to overlay with multiple grid cells; as shown in Fig. 4, the road segment with ID 110,577 couples with both grid cells 16,089 and 15,789, but these two grid cells have different PM_{2.5} concentrations. To accurately measure the relative exposure risk to PM_{2.5}, road segment 110,577 needs to be separated at the intersection point of the road segment and grid lines, which can be achieved by using the intersect tool in ArcGIS.

After breaking the road network segments according to the intersections with the grid lines, the relative exposure level of each road segment can be calculated based on the PM_{2.5} value of the grid cell that its couples with, and the total relative exposure level of PM_{2.5} can be derived by adding all the involved road segments together. The relative exposure level of a road segment at a given time can be calculated as follows:

$$R_i^{m,t} = \frac{d_i^m * P_i^t}{v^m * h_0 * P_0} * r_0 \quad (8)$$

where $R_i^{m,t}$ represents the relative exposure level of PM_{2.5} of road segment m in grid cell i at time t , and d_i^m represents the length of road segment m within grid cell i . v^m is a term representing the speed of the user passing road segment m ; users can choose either biking or walking. P_i^t represents the average level of PM_{2.5} of grid cell i at time t . P_0 is a constant value representing the rated standard level of PM_{2.5}, determined by the WHO's suggested safe level of PM_{2.5} (no negative impact on health when below this value), which is $15 \mu\text{g}/\text{m}^3$ on average within 24 h (World Health Organization, 2021). h_0 is a constant term that represents the rated exposure time, which is set to 24 h as a rated value since P_0 is calculated based on a 24-hour average. r_0 is a constant magnification factor, which is set to 100 so that the relative exposure level can appear to be more significant. Finally, we can evaluate the total exposure to PM_{2.5} for the whole route.

The route with the least exposure to PM_{2.5} can be queried in Neo4j after updating each road segment with its relative PM_{2.5} exposure rate. The command used to update the original properties of each road network segment is "MATCH (node1 {Node_ID: Num1})-[r]-(node2 {Node_ID: Num2}) SET r={Weight: Risk}". Here, Num1 and Num2 represent the IDs of the two nodes, r represents the relationship between the two nodes, and Risk represents the calculated relative exposure level of PM_{2.5} for that road segment. After the weight properties are updated with the relative PM_{2.5} exposure levels, the route between two nodes with the least exposure to PM_{2.5} can be queried with the graph algorithm provided by Neo4j. The algorithm calculates the least traversal between nodes based on the minimum combined weight of the edges involved in the route. The implementation of the algorithm is supported

by a Neo4j plug-in called APOC, which contains abundant processes and functions related to graph data query, integration, and management (Kemper, 2015). The command that calls the Dijkstra graph algorithm to query the route with the least PM_{2.5} exposure is "apoc.algo.dijkstra(startNode, endNode, '<Knows>', 'distance') YIELD path, weight". Here, 'apoc.algo.dijkstra()' is the shortest route selection algorithm provided by APOC, 'startNode' and 'endNode' represent the starting point and destination of the query, '<knows>' represents the property label of the relationship, and distance means that the query will be based on that property of the relationships. This query will return the path of the selected route and the corresponding weights.

4. Case of the UAHNS

4.1. Study area

To demonstrate the proposed Urban Air Health Navigation System, we chose Wuhan, the largest city in central China, as our example urban area. Wuhan spans over an area of 8,569 km², and it comprises 13 administrative districts. The city has a permanent population of 13.65 million, and its rapid urbanization over the years has led to a range of atmospheric environment issues while promoting economic growth (Zhang et al., 2024). PM_{2.5} caused by traffic emissions is an important component of air pollutants in Wuhan, and it has a negative impact on the health of residents (Deng et al., 2024; Yang et al., 2020). As a typical mega-city in China, Wuhan has a relatively large built-up area divided by rivers, complicated land use, and frequent commercial flow. A large number of residents in Wuhan need to use bicycles or e-bikes to commute medium or long distances during rush hour (Zhang et al., 2022), which further increases the amount of PM_{2.5} inhaled and the associated health risks.

4.2. System functional analysis

The user application is an essential part of the proposed UAHNS, as it is the component that users directly interact with. System requirements need to be defined and clarified so that the system structure can be designed around them to provide a better user experience. The system first needs to satisfy the functional requirements for users, and performance requirements such as a fast response time and an intuitive user interface must also be satisfied to ensure a pleasing user experience. The functional requirements are detailed below.

Route query serves as the core functionality of the UAHNS. It should efficiently plan the route with the least PM_{2.5} exposure for users according to their given starting point and destination. The queried route in Neo4j needs to be returned to the front-end browser and visualized through the map API. At the same time, when users query the route, it should return both the route with the least exposure to PM_{2.5} and the route with the shortest distance, along with its relative exposure to PM_{2.5}, so that users can also consider time and distance costs when planning their commute.

To further enhance the system's interactivity and users' awareness of their surrounding environment, the user application should incorporate the ability to display the latest PM_{2.5} concentrations from monitors across the urban area. By left-clicking a specific location on the map, users can obtain both the current and estimated PM_{2.5} concentrations for the next hour, providing a more accurate understanding of the air quality in that area. Additionally, to help users appreciate the practical significance of PM_{2.5} concentrations, we incorporated an exposure standard module. This module displays the latest international standards (2021 version) from the World Health Organization (WHO) and the most recent Chinese standards based on GB3095-2012, allowing users to reference and compare PM_{2.5} levels against these benchmarks. Each standard can be easily accessed through hyperlinks.

Furthermore, to assist users in understanding their PM_{2.5} exposure during navigation with the UAHNS, we introduced a "Personal Center"

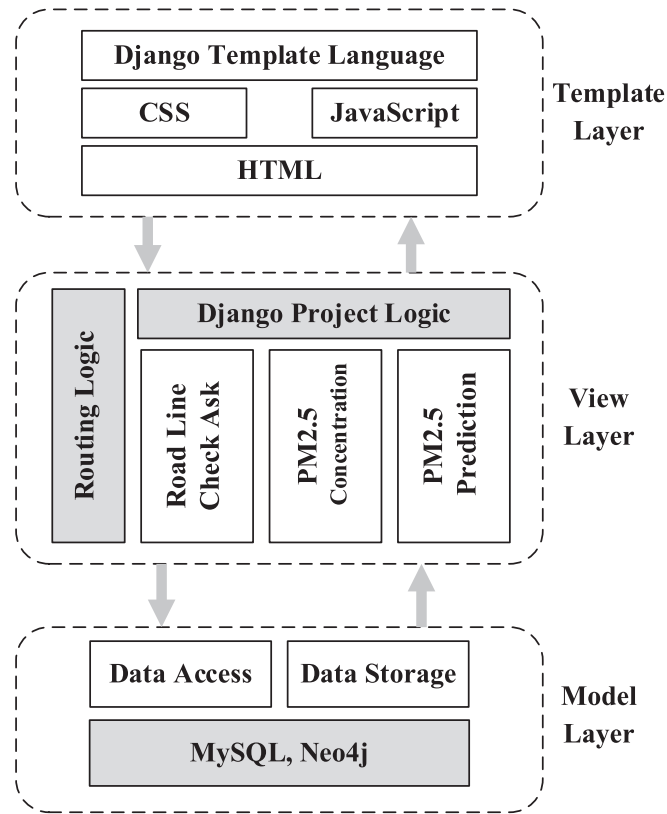


Fig. 5. Three-layer structure of UAHNS.

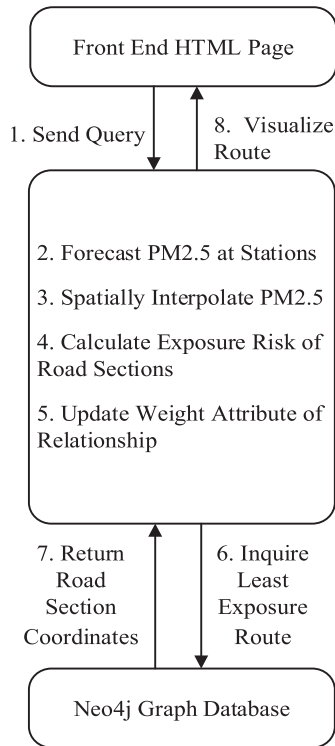


Fig. 6. Working logic of route query.

page. Here, users can input personal information, including their gender, age, height, and weight. This page also allows users to query their personal daily exposure to PM2.5 during navigation. The exposure

Table 2
Optimized hyperparameters.

Parameter name	Parameter interpretation	Parameter optimal value
n_estimators	Number of subtrees	200
learning_rate	Step size for weight updates	0.05
max_depth	Maximum tree depth for decisions	6
min_child_weight	Minimum sum of instance weights for a child node	2
reg_alpha	L1 regularization strength	0
reg_lambda	L2 regularization strength	1.50

assessment algorithm utilized in the background calculates an individual's level of PM2.5 exposure by combining their personal breathing volume data from the gender and age query table with the PM2.5 concentration data obtained through spatial interpolation and the duration of navigation usage.

4.3. User application system structure design

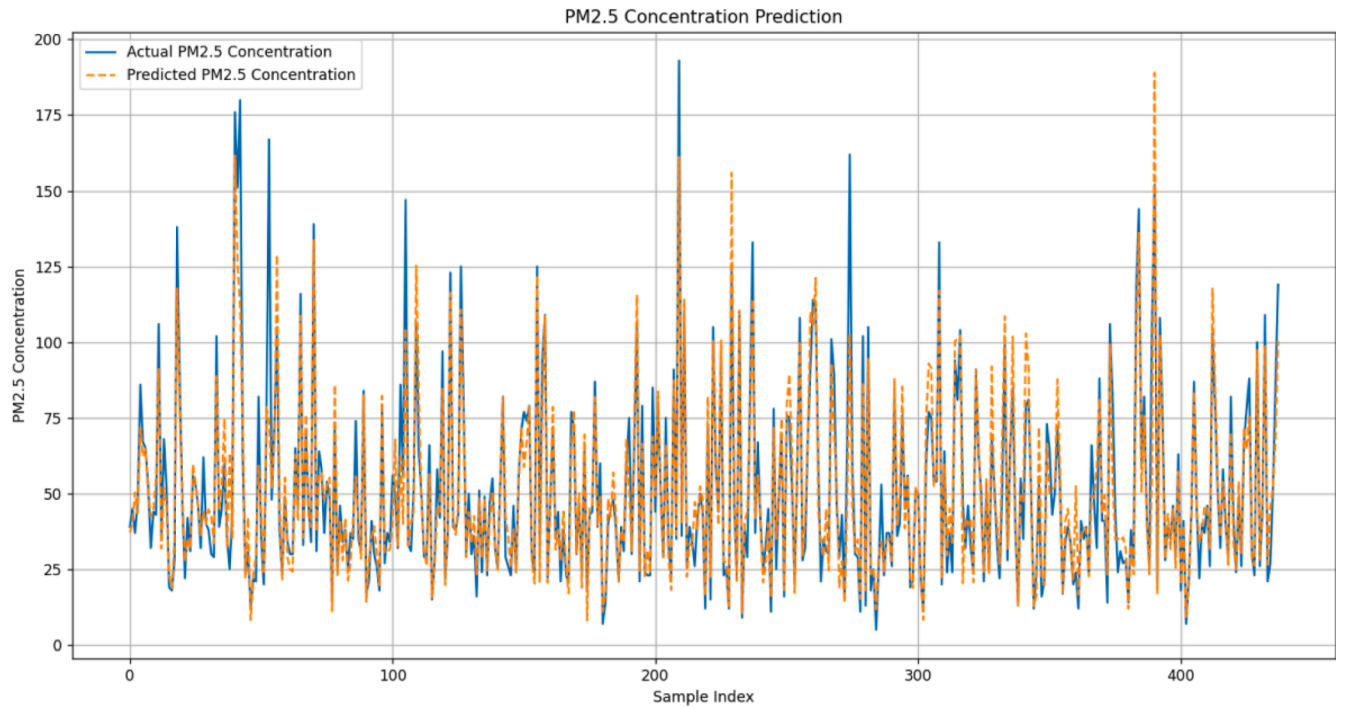
The UAHNS is designed based on the browser/server pattern according to the system requirements and using the Django framework (Hu, 2019). HTTP requests are sent from the user side to the server from the browser, and the requests are transferred to the Django project through the back-end router. The completed logic processed in Django is returned to the front-end browser in JSON format and then visualized.

The Django framework consists of a model layer, template layer, and view layer. It combines the function of front-end visualization with the decoupling of back-end logic processing and data models, facilitating the development of system modules and operational maintenance. The model layer serves the view layer through data transfer and mapping, maintaining the object and database relationship. It supports the storage and query of road network segment, air quality, and meteorology data. Since Neo4j has high performance in storing and querying road networks and can efficiently call the graph algorithm in the storage process, it is used as the database. The template layer dynamically renders the HTML web page for the front-end browser by using template language to load the returned data from the view layer to template files, providing a human-computer interface. It serves as a middle layer that communicates between the users and the back-end system. The view layer serves as the core layer in our user application. It responds to users' requests by calling the model layer for data processing and feeds the result to the template layer to render the HTML web page. The overall structure of the UAHNS user application is shown in Fig. 5.

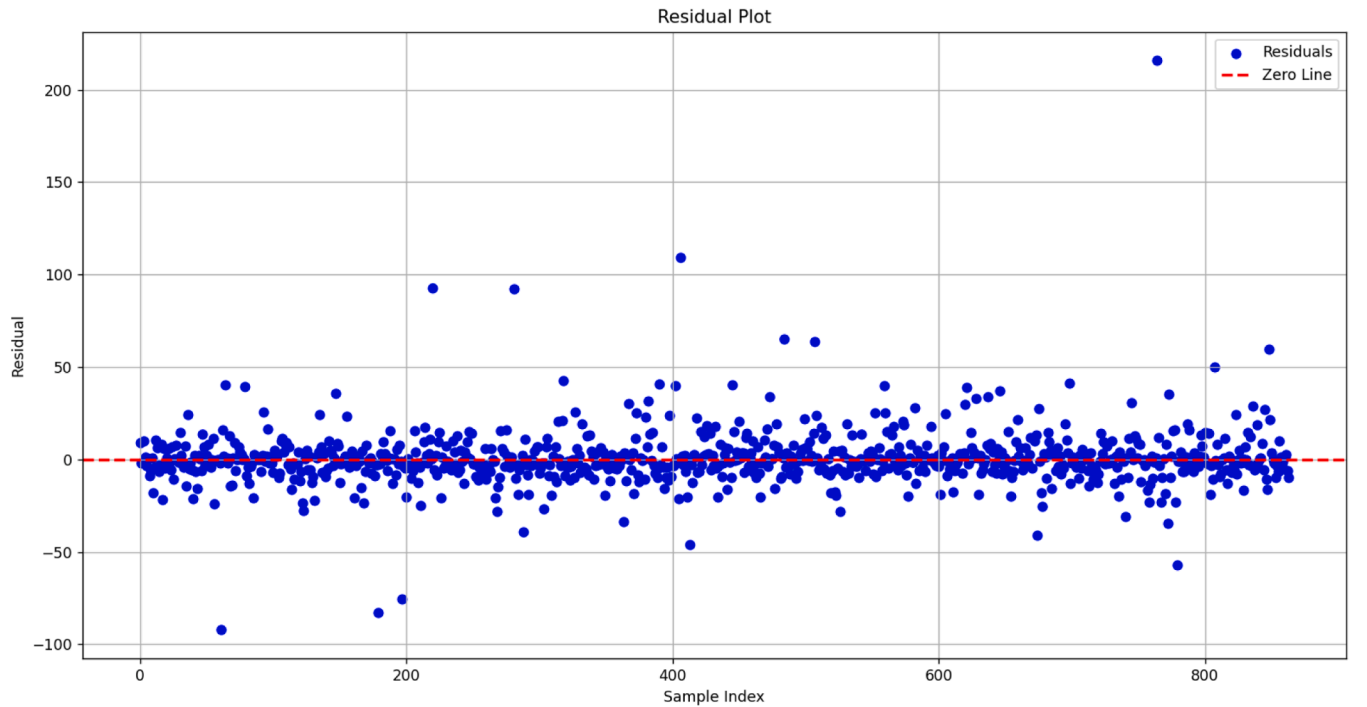
4.4. Route query model

This model handles the route query and analysis in response to users' requests; it includes the selection of the routes with the least PM2.5 exposure and with the least distance. Information on the relative exposure to PM2.5 and the selected route's total length is displayed. Its query logic for commuting in one hour is shown in Fig. 6.

The query for the route with the least PM2.5 exposure operates as follows: Initially, users input the GPS coordinates of their starting point and destination, which serve as the query parameters. These parameters are then sent to the back-end, initiating the Django framework's project logic. This logic begins by predicting the PM2.5 concentrations for each air quality station in the next hour. Subsequently, the two-step spatial interpolation method is employed to determine the PM2.5 distribution across the urban area. Later, the relative exposure risk of each road segment to PM2.5 is calculated to update each relationship's weight in the Neo4j graph database. In this case, the least weighted route, which is the route with the least PM2.5 exposure, will be queried, and the result will be returned to the back-end server. Finally, the front-end web page will receive the returned set of GPS coordinates of the route, which will be visualized using the Jason web map plugin. The shortest route query



(a)



(b)

Fig. 7. Training results: (a) a comparison between the predicted and actual results and (b) the residual of the test set.

is an option for users when considering the distance and time costs of travel. It only uses the length of a road segment as the weight of a relationship.

5. Testbed for UAHNS

5.1. PM2.5 prediction test in Wuhan

Hourly meteorological and air pollutant monitoring data from Wuhan in 2023 are used. They are all normalized and used as the input

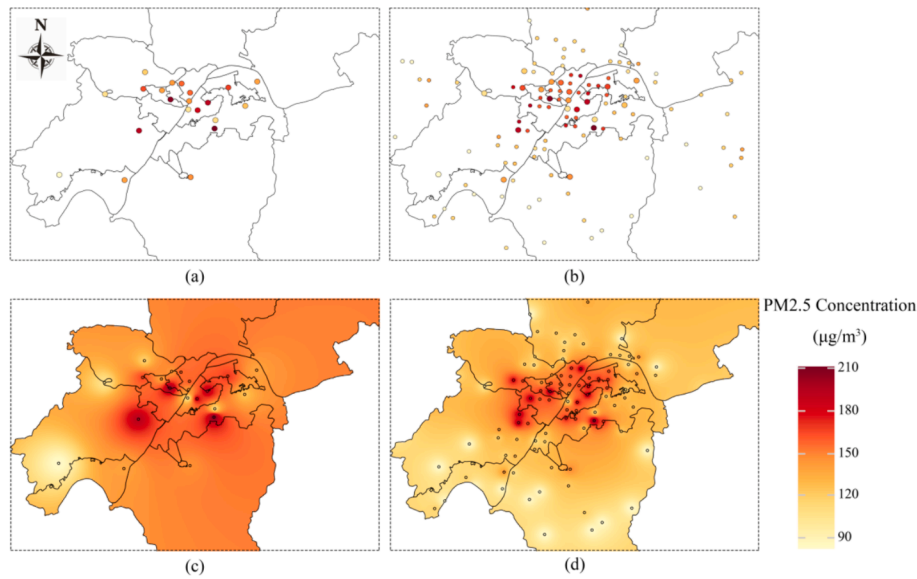


Fig. 8. Schematic diagram of the interpolation process and results. (a) PM2.5 concentrations at 21 monitoring stations. (b) After data augmentation based on environmental similarity, PM2.5 at certain grid points is predicted and incorporated into our dataset. (c) The result of IDW interpolation using only the 21 monitoring stations from (a). (d) The result of IDW interpolation based on the data from (b).

Table 3
Quantitative comparison between the random selected travel scenarios.

Commuting mode	Route with least distance		Route with least exposure		Route difference ratio (%)
	Distance (km)	Exposure risk	Distance (km)	Exposure risk	
Riding	13.80	23.34	17.10	19.86	14.90
Walking	4.10	7.34	5.87	6.10	16.90

features to the XGBoost models. The number of decision trees ($n_{estimators}$), the learning rate ($learning_rate$), and the maximum depth of the trees (max_depth) are the three main hyperparameters that affect the performance of the XGBoost model. In addition to these, the minimum child weight of each leaf node (min_child_weight) and L1 and L2 regularizations are also included. Subsequently, GridSearchCV is used to perform grid search cross-validation on the XGBoost regression model XGBRegressor, with a five-fold cross-validation (the ratio of the training set to the test set is randomly divided at a 8:2 ratio), using the root mean squared error ($root_mean_squared_error$) as the scoring function to obtain the optimal combination of parameters and achieve better predictive results. The identified optimal parameters are shown in Table 2, with an R^2 exceeding 0.90, an RMSE of approximately $15.74\ \mu\text{g}/\text{m}^3$, and an MAE of $8.74\ \mu\text{g}/\text{m}^3$, demonstrating its good applicability in the present study.

The blue solid line in Fig. 7(a) represents the actual PM2.5 concentration values, while the orange dashed line represents the PM2.5 concentration values predicted by the model. It can be observed in the chart that the predicted values generally follow a similar trend to the actual values, indicating that the model can effectively capture the pattern of PM2.5 concentration changes. The Fig. 7(b) shows that most of the points are clustered around the zero line, suggesting that the prediction errors are within an acceptable range. After the XGBoost model is trained, the latest one-hour average air pollutant monitoring and meteorological data are used to predict the PM2.5 in next hour.

5.2. Realization and verification of selection of route with least exposure

The interpolation map of PM2.5 distribution at a spatial resolution of $300\text{ m} \times 300\text{ m}$, which are constructed from 120 sampling points based on

environmental similarity (Fig. 8(b)), is presented in the Fig. 8 (d). For comparison, a PM2.5 interpolation map was also created using only the 21 monitoring stations (Fig. 8(a) and (c)). The comparison maps were shown in, which clearly demonstrated that the environmental similarity method yielded a more actual and high-resolution PM2.5 distribution.

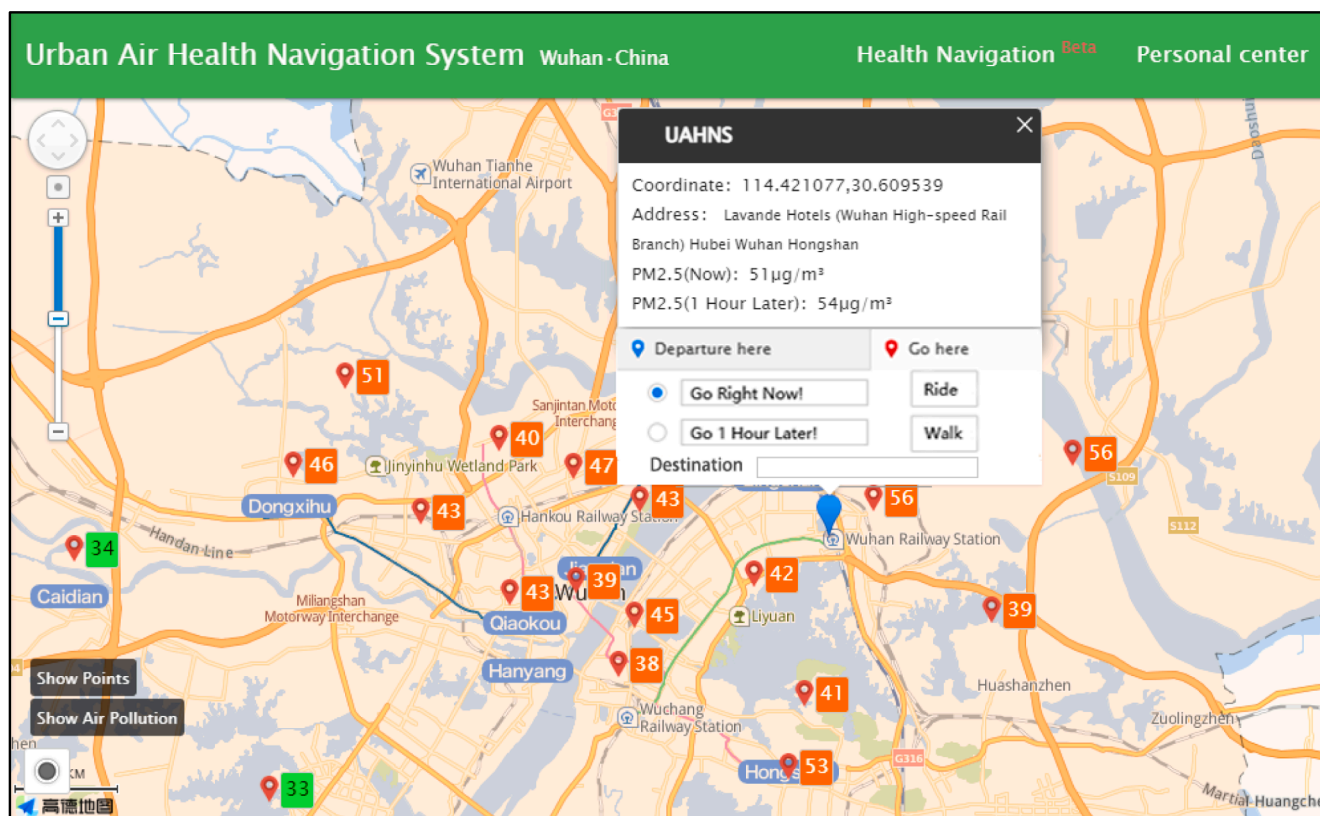
To demonstrate the effectiveness of the Urban Air Health Navigation System in reducing users' exposure to PM2.5 during commuting, a quantitative comparison of relative PM2.5 exposure between taking the system-suggested route and the route with the shortest distance was conducted. Two common user travel scenarios were randomly established, which are traveling by riding and walking. Then, the shortest routes and the routes with the least PM2.5 exposure were queried in UAHNS, and the quantitative comparison results that system returns are shown in Table 3.

As shown in Table 3, although the route with the least PM2.5 exposure requires a longer travel distance (about 20 %) and time, it effectively reduces the risk of PM2.5 exposure (about 15 %). When users travel using the route with the least exposure, the PM2.5 pollution level is lower, and the exposure risk is reduced by 14.9 % and 16.9 % when cycling and walking, respectively. Further, more travel scenarios were tested to demonstrate the effectiveness of the UAHNS, and the results could be found in Table A2 in Appendix.

5.3. UAHNS platform deployed for Wuhan

Fig. 9 (a) presents the home page of the Urban Air Health Navigation System User Application, serving as the primary entry point for users to engage and interact with the web application. The homepage comprises three distinct sections: navigation menus situated at the top, a prominent web map visualized using a map API, and various manipulation buttons for user convenience.

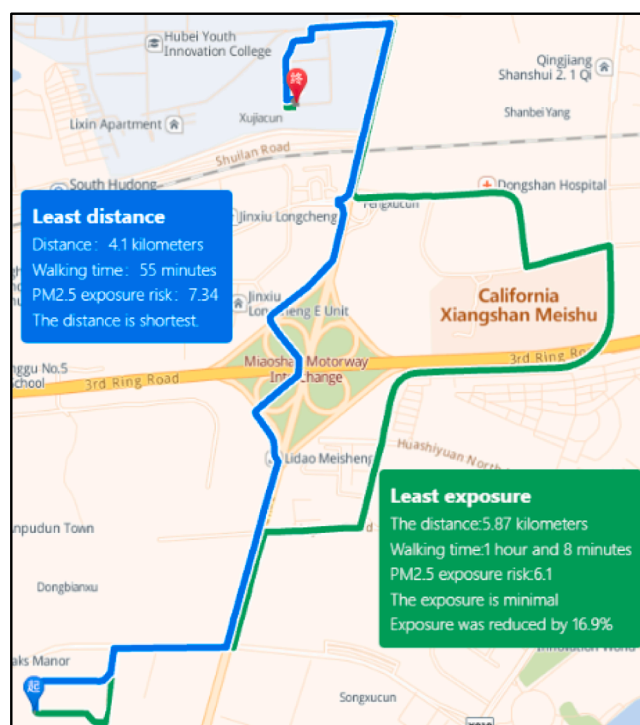
The home page shows a map of Wuhan, where the air monitoring sites were also shown with their PM2.5 concentrations. The most recent PM2.5 concentration monitoring values are presented to the right of each monitoring site, with varying background colors that adhere to international and national standards, providing a clear and intuitive representation of air quality levels. When user left-clicks the web map to select the start point or destination, then an information box pops up. This box presents pertinent details such as coordinates, address, current PM2.5 concentrations and their predicted concentrations for the



(a)

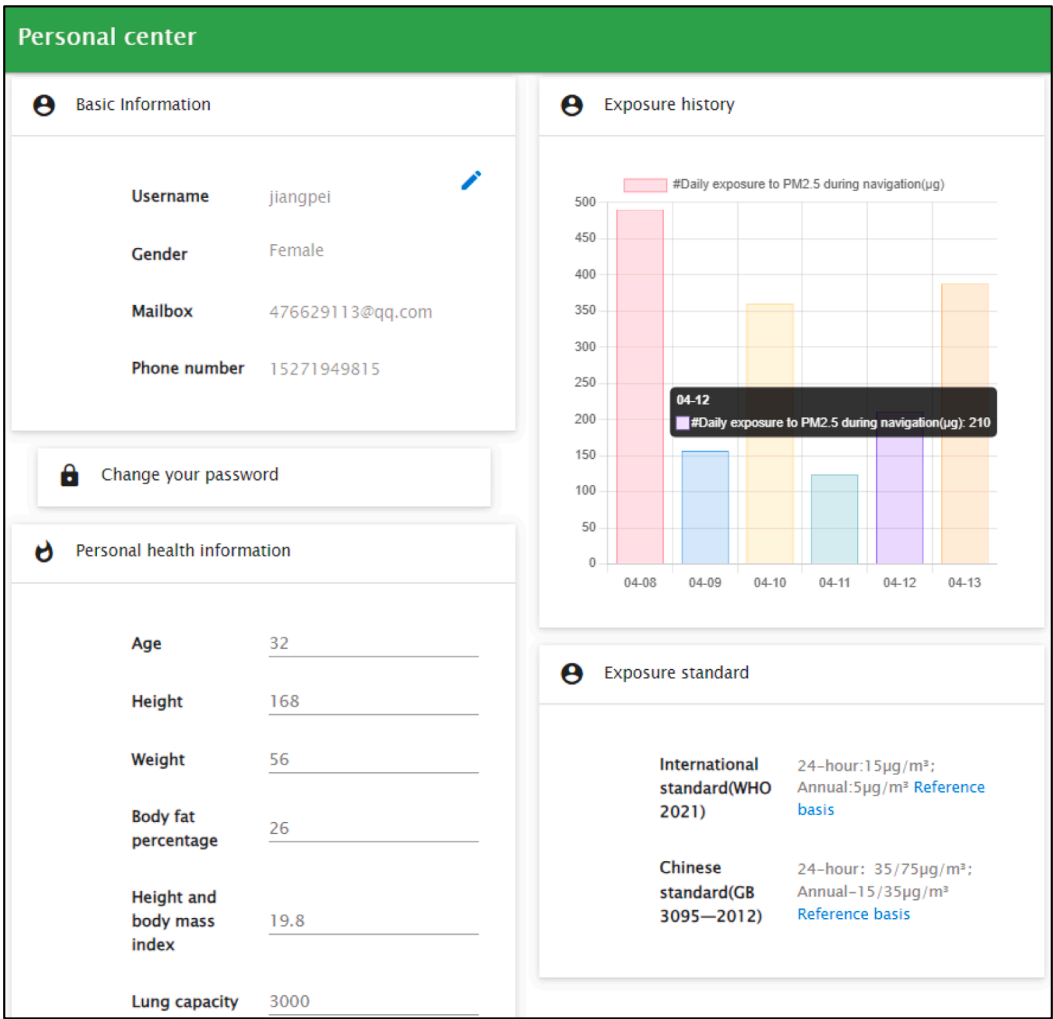


(b)



(c)

Fig. 9. View and function display of UAHNS: (a) index page, (b) riding route query result, (c) walking route query result, (d) page of personal center.



(d)

Fig. 9. (continued).

following hour. Notably, the current PM2.5 concentration at any position is determined using the two-step spatial interpolation method, while the prediction concentration for the next hour relies on results by the established XGBoost algorithm.

Crucially, the system enables users to input their desired destination or start point. Firstly, users can choose “Go Right Now” or “Go 1 Hour Later”, and then they can click on the “walk” or “ride” query button once the start point and destination are set. Once the back-end completes the processing of the route query, it will send the set of coordinates of the route to the web page and draw the route on the web map using JavaScript. Additionally, the related information windows will appear, displaying both the routes with the shortest distance and with the least PM2.5 exposure. Adjacent to the displayed route on the page, the system also indicates the length of the route, the estimated travel time, and the amount of exposure. This allows users to clearly discern the differences between the two routes and subsequently choose the one that best suits their needs. Fig. 9 (b) and (c) illustrate the examples of the queried routes.

The main functional modules can be accessed by clicking the tabs on the top of the web page. Inside of the navigation bar area, “Personal center” is noteworthy. Fig. 9 (d) provides a snapshot of this personal center page. In addition to providing the functions of registration and login, this system also has the function of the personal center entry of basic information, including mailbox, phone number, gender, age,

height, weight, and lung capacity. Furthermore, on this page, users could also check their daily statistical data of accumulated PM2.5 exposure quantity after their navigation service usages. The exposure assessment algorithm applied in the background calculates personal PM2.5 exposure based on the user’s personal breathing volume data in the gender and age query table, combined with the PM2.5 concentration data of the two-step spatial interpolation method and the time of the navigation. To assist users in understanding the practical significance of the exposure quantity, there is an international exposure standard module on the “Personal center” page. This module displays the limited values based on the latest standards from the World Health Organization (WHO) and Ministry of Ecology and Environment of the People’s Republic of China (GB3095-2012), providing users with key reference baselines. Additionally, each standard has a hyperlink providing access to the corresponding standard’s text for further reading.

6. Conclusions

In this study, based on topologized digitized maps, an application programming interface (API), a trained XGBoost model, and two-step spatial interpolation, the UAHNS framework is proposed and established to assist users by recommending the route with the least exposure and dynamically issuing early risk warnings. A testbed with the UAHNS functions and applications was accomplished in Wuhan. Based on the

data from national air and meteorological monitoring stations, the trained XGBoost model performed well, achieving an R^2 of at least 0.90, indicating a good prediction ability for ambient PM2.5 and city-level scalability. Additionally, the two-step spatial interpolation method was adopted to dynamically generate pollution distribution at a spatial resolution of 300 m*300 m. Further, the commuting routes and times were randomly selected in Wuhan, and a comparison was performed, with the results showing that routes with the least PM2.5 exposure could effectively help in reducing the PM2.5 exposure risk, with route difference ratios of about 14.9 % and 16.9 % for riding and walking, respectively. Facing demands for a low-carbon and healthy lifestyle, the UAHNS platform was integrally realized for Wuhan to assist users in making decisions regarding when and how to commute, engage in outdoor fitness, and use low-carbon mobility (ride a bike, jog, walk, etc.). The UAHNS is a promising system framework, although it is currently being explored as a preliminary system, with lots of work required in future research. A broader range of urban commuters, regional high-precision indoor/outdoor air pollutant simulations with IoT sensors, personal health economic assessments and management, technology for protecting user privacy, user-level carbon smart accounting and corresponding carbon inclusive finance, etc., need to be continuously explored and integrated into the UAHNS.

Availability of data and materials

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Appendix. . Additional Table and illustration

Table A1

The short-term inhalation rates of Chinese residents*.

Factor	Category	Number of surveyed population (person)	Sedentary	Light intensity activity	Moderate intensity activity	High intensity activity	Extremely high intensity activity
			Median (L/min)				
Sex	Male	41,261	7.5	9.4	25.1	37.7	62.8
	Female	49,798	6.1	7.6	20.3	30.4	50.7
Age	18–44	36,662	6.9	8.6	23.0	34.5	57.6
	44–59	32,352	6.9	8.6	22.9	34.4	57.3
	60–79	20,561	5.7	7.1	19.0	28.5	47.4
	≥80	1,484	5.0	6.2	16.5	24.8	41.4

* Note: Liu P, Wang BB, Zhao XG, et al. (2014) Research on inhalation rate of Chinese adults. J ENVIRON HEALT 31(11): 953–956. (in Chinese).

Table A2

Quantitative comparison results of different routes under various scenarios.

Route inquiry area	Route inquiry time	Route with least distance		Route with least exposure		Route difference ratio (%)
		Average distance (km)	Average risk	Average distance (km)	Average risk	
Low concentration area to high concentration area	9 a.m.	39.362	65.865	60.253	54.237	21.439
	8p.m.	48.981	79.790	55.721	63.632	25.393
	Total	88.343	145.655	115.974	117.869	23.574
High concentration area to high concentration area	9 a.m.	49.063	97.444	54.574	95.913	1.596
	8p.m.	54.315	107.642	61.542	106.472	1.099
	Total	103.378	205.086	116.116	202.385	1.335
Low concentration area to low concentration area	9 a.m.	37.104	19.601	44.585	18.532	5.768
	8p.m.	34.872	17.960	40.957	16.044	11.942
	Total	71.976	37.561	85.542	34.576	8.633

CRedit authorship contribution statement

Pei Jiang: Writing – original draft, Methodology. **Chang Gao:** Writing – original draft. **Junrui Zhao:** Writing – review & editing, Data curation. **Fei Li:** Writing – review & editing, Project administration, Methodology. **Changhong Ou:** Writing – review & editing, Data curation. **Tao Zhang:** Methodology, Formal analysis. **Sheng Huang:** Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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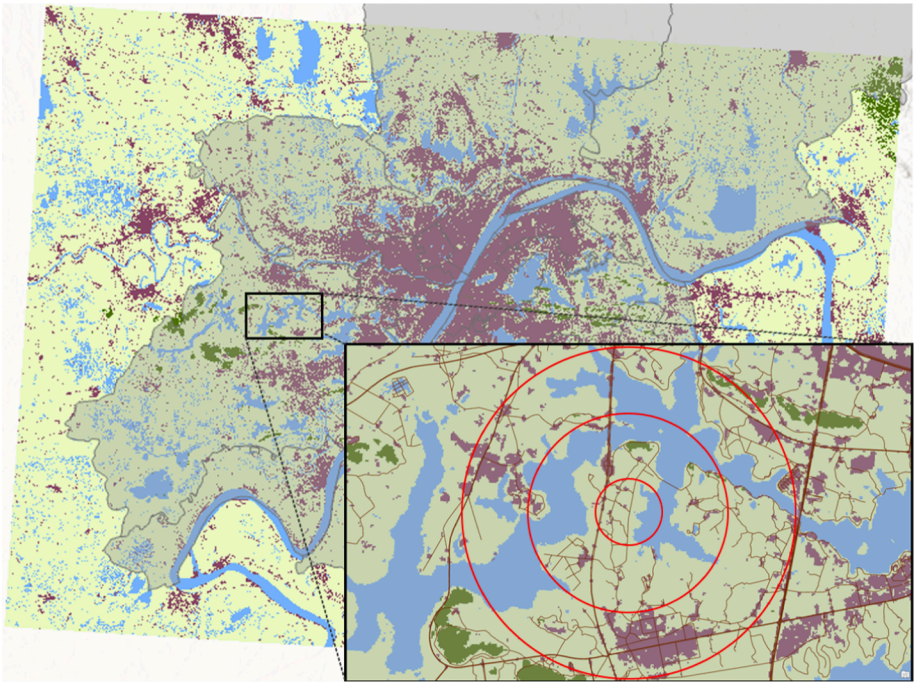


Fig. A1. Study area, land use data, and environment feature buffer (300 m, 900 m, and 1500 m) of one point to be estimated.

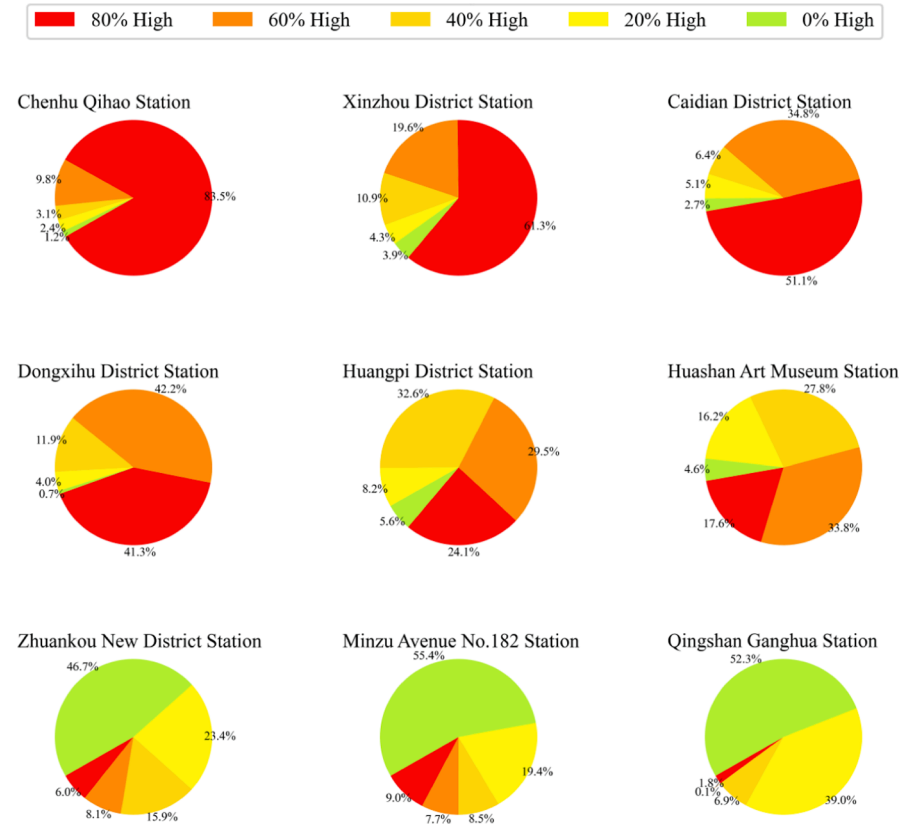


Fig. A2. Statistical ranking results of pollution levels for 9 monitoring stations.

The red color labeled “80 % High” indicates at a certain time, the pollution value of this station exceeds 80 % in all ones. The proportionate size of the red segment in the pie chart represents the percentage of time that this condition persists throughout 2023. The orange/ light orange/ yellow/ light green color present the results of “60 % High”/ “40 % High”/ “20 % High”/ “0% High”. For example, “0% High” means this station is one of the 20 % least polluted ones in all stations at that time.

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